

# VPA ICT Program — Grades 4–5 Course Overview

## VPA ICT Program — Grades 4–5: Digital Fluency

Academic Year: 2024 – 2025 | Total Weeks: 36 | Lesson Duration: 40 minutes

Target Group: EAL learners (Mandarin L1), mixed ability | Standards: CSTA K–12, ISTE Standards

Typing Target: 20 WPM (words per minute) by end of Grade 5 with 90% accuracy

### Program Vision

Grades 4–5 build on digital foundations by developing fluency in productivity tools (word processing, spreadsheets, presentations), responsible internet use, and introductory programming with Scratch. Students begin creating polished digital products for real audiences while developing critical thinking about online information.

### Curriculum Map — Three Terms

| Term   | Weeks   | Theme                        | Key Skills                                                  | Standards                                   |
|--------|---------|------------------------------|-------------------------------------------------------------|---------------------------------------------|
| Term 1 | 1 – 12  | Productivity Tools           | Advanced word processing, formatting, images, tables        | CSTA 1B-DA-06, ISTE 1.6, CCSS.ELA.W.4.6     |
| Term 2 | 13 – 24 | Spreadsheets & Presentations | Excel basics, formulas, charts, PowerPoint                  | CSTA 1B-DA-07, ISTE 1.5, CCSS.MATH.4.OA.C.5 |
| Term 3 | 25 – 36 | Research & Coding            | Internet research, Scratch programming, digital citizenship | CSTA 1B-AP-10, ISTE 2.1, CCSS.ELA.W.4.7     |

### Year at a Glance

| Week | Topic                          | Key Activity                      |
|------|--------------------------------|-----------------------------------|
| 1    | Typing Review & Goal Setting   | Baseline typing test (WPM)        |
| 2    | Touch Typing Practice          | TypingClub Level 1 – 3            |
| 3    | Speed & Accuracy               | Typing race tournaments           |
| 4    | Advanced Formatting            | Professional document challenge   |
| 5    | Tables in Documents            | Class timetable creation          |
| 6    | Images & Wrapping              | Travel brochure project           |
| 7    | Headers, Footers, Page Numbers | Create a formal letter            |
| 8    | Review & Mini-Project          | "All About Me" formatted document |
| 9    | Document Collaboration         | Shared Google Doc editing         |
| 10   | Proofreading & Editing         | Error-finding challenge           |
| 11   | Real-World Writing             | Write a letter to the principal   |

| Week | Topic                         | Key Activity                     |
|------|-------------------------------|----------------------------------|
| 12   | Term 1 Assessment             | Document production test         |
| 13   | Spreadsheet Introduction      | Excel/Sheets exploration         |
| 14   | Data Entry & Formatting       | Class birthday chart             |
| 15   | Basic Formulas (+, -, *)      | Shopping list calculator         |
| 16   | AVERAGE, MAX, MIN             | Sports score analysis            |
| 17   | Creating Charts               | Bar chart of favorite animals    |
| 18   | Pie & Line Charts             | Weather data visualization       |
| 19   | Introduction to Presentations | PowerPoint/Slides basics         |
| 20   | Slide Design & Layouts        | Consistent theme challenge       |
| 21   | Animations & Transitions      | "Do's and Don'ts" of animations  |
| 22   | Embedding Media               | Presentation with video          |
| 23   | Presenting Skills             | Peer presentation practice       |
| 24   | Term 2 Assessment             | Data + Presentation project      |
| 25   | Internet Research Skills      | Keyword search strategies        |
| 26   | Evaluating Sources            | Real or Fake? website game       |
| 27   | Email Etiquette               | Write a professional email       |
| 28   | Digital Footprint             | "What will Google say about me?" |
| 29   | Introduction to Scratch       | Sprites, backdrops, blocks       |
| 30   | Movement & Events             | Maze game project                |
| 31   | Loops & Variables             | Score-keeping game               |
| 32   | If/Then & Sensing             | Quiz game with scoring           |
| 33   | Sprites & Costumes            | Animated story project           |
| 34   | Sounds & Broadcasting         | Interactive story                |
| 35   | Review & Debug                | Bug-fixing challenge             |
| 36   | End of Year Assessment        | Portfolio + coding project       |

## Assessment Overview

Formative (ongoing):

Weekly typing speed checks (WPM tracking chart)

Document / spreadsheet / presentation checkpoints

Scratch project milestones (movement → loops → variables)

Peer code reviews with structured feedback forms

Summative (term-end):

Term 1: Timed document production task (40 min)

Term 2: Data analysis + presentation project

Term 3: Research report + Scratch game project

## Bilingual Support

Sentence Frames for Writing:

"The data shows that..." / "数据显示... (shùjù xiǎnshì...)"

"According to my research..." / "根据我的研究... (gēnjù wǒ de yánjiū...)"

"I recommend... because..." / "我建议...因为... (wǒ jiànyì... yīnwèi...)"

## Differentiation

| Level  | Word Processing                               | Spreadsheets                                   | Coding                                |
|--------|-----------------------------------------------|------------------------------------------------|---------------------------------------|
| Bronze | Uses templates with fill-in boxes             | Enters data into pre-made tables               | Follows step-by-step video tutorial   |
| Silver | Creates documents from scratch with checklist | Creates formulas with scaffolding              | Adapts example projects               |
| Gold   | Creates complex multi-page documents          | Designs own spreadsheet with multiple formulas | Creates original projects from design |